

Response to Reviewer Comments

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability
via Graph Neural Networks

ACM-BCB 2026 Submission

We thank all three reviewers for their thorough and constructive feedback. We have substantially revised the manuscript, and all changes are [highlighted in blue](#) in the revised paper. Line numbers below refer to the revised `paper.tex`. The major revisions include:

- **6-seed extended validation** (seeds 42, 123, 456, 789, 1011, 1213): the 6-seed mean is 0.603 ± 0.097 , honestly reported alongside the original 3-seed mean (0.646 ± 0.124).
- **Statistical significance testing**: bootstrap p -values confirm the improvement over gradient boosting is not significant ($p = 0.259$ for 3-seed, $p = 0.556$ for 6-seed). This is stated explicitly in the main text.
- **VHL root cause analysis**: we rule out class imbalance and dataset size, identifying structural heterogeneity as the likely cause of VHL failure (0.396 AUROC).
- **BRD4 biological case study**: lysine attention weights identify K374 as the top-weighted residue, validated by mass spectrometry literature.
- **Calibration analysis**: $\text{ECE} = 0.029$ on target-unseen (excellent), providing a new quantitative strength.
- **Narrowed claims**: E3 generalization narrowed to $\text{CRBN} \leftrightarrow \text{VHL}$ transfer; multi-seed average is the primary reported result; limitations stated prominently before results.

Response to Reviewer 9v2E (Rating: 4 — Borderline Accept)

C1: The dataset is relatively small (3,101 samples, 155 unique proteins, 10 E3 ligases). The target-unseen effective diversity is much closer to the number of unique proteins, contributing to seed sensitivity.
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Response: We fully agree. We have added a prominent “Dataset characteristics and limitations” paragraph *before* any results are presented:

“PROTAC-8K contains 3,101 labeled samples spanning only 155 unique protein targets... The E3 ligase distribution is highly imbalanced: CRBN accounts for 62% of samples, VHL for 35%, and the remaining eight E3s combined represent only 3%. These constraints fundamentally bound generalization performance...”

We also note that PROTAC-8K is the only publicly available benchmark of this scale, and we have no external validation dataset—both stated as explicit limitations.

[\[Lines 431–436 \(new paragraph before results\); Limitations items 1, 2, 6 at lines 766, 767, 771.\]](#)

C2: Structural validation is limited. Predictions are not supported by docking, ternary-complex modeling, MD, or free energy calculations.

Response: We agree this is a limitation. It is now discussed in the Limitations paragraph:

“Predictions rely on static AlphaFold structures without physics-based validation (docking, MD, free energy calculations). The BRD4 case study (K374) provides one anecdotal positive control.”

We note an inherent tension: ternary-complex modeling requires knowing the PROTAC linker geometry, which is unavailable in our pre-synthesis setting. Post-hoc structural validation against experimentally determined ternary complexes (where available) is an important future direction.

[Limitation item 5 at line 770.]

C3: No compelling biological case study showing why a specific protein is predicted degradable, which lysines drive the prediction, or how predictions align with literature.

Response: We have added a new subsection “Case Study: BRD4 Degradability” that provides exactly this analysis:

- BRD4 (O60885) is predicted highly degradable with CRBN (score: 0.87).
- Lysine attention weights identify **K374** (attention: 0.12, SASA: 94 Å²) as the top residue—located in a disordered, solvent-exposed linker between bromodomains.
- Literature validation: mass spectrometry studies by Zengerle et al. (2015) confirm K374 as a major ubiquitination site following JQ1-based PROTAC treatment.

We acknowledge this is a single successful case and that systematic validation across more targets would strengthen confidence.

[Lines 686–703 (new §4.5 “Case Study: BRD4 Degradability”).]

Additional: The key target-unseen result is unstable; the paper emphasizes the best seed over the multi-seed result; some biological and E3 generalization claims are only partly supported.

Response: All three points are now addressed:

1. **Instability:** 6-seed extended validation (0.603 ± 0.097) reported alongside the original 3-seed (0.646 ± 0.124); ensembling across ≥ 6 seeds is explicitly recommended for deployment (lines 619–625).
2. **Best seed de-emphasized:** the multi-seed average is the primary result throughout; best seed (0.7449) is consistently marked as secondary, parenthetical (Abstract line 65; Table 1 at line 457; Conclusion line 783).
3. **E3 claims narrowed:** changed throughout from “E3 generalization” to “CRBN→VHL transfer” (Abstract line 65; section heading line 635; Conclusion line 783); rare E3 generalization explicitly stated as unsupported (line 648).

Response to Reviewer Y2rf (Rating: 2 — Reject)

We appreciate the reviewer’s detailed and rigorous critique. We have addressed every major and minor concern and believe the revised manuscript warrants reconsideration.

M1: The main result should be multi-seed average, not best seed. Rewrite abstract, results, and conclusion accordingly.

Response: Done. Throughout the revised manuscript:

- The abstract leads with “ 0.646 ± 0.124 AUROC on target-unseen evaluation (best seed: 0.7449)” (line 65).
- The introduction reports both 3-seed and 6-seed means as primary numbers (lines 135–136).
- Table 1 (line 444) now reports both 3-seed (0.646) and 6-seed (0.603) means, with “Best Seed” as a separate column.
- Section 4.2 states: “We report multi-seed averages as primary results... best-seed performance (0.7449) represents peak achievable performance but should not be interpreted as typical expectation” (lines 458–461).
- The conclusion leads with average performance (line 783).

[Lines 65, 135–136, 444–456 (Table 1), 458–461, 783–785.]

M2: Target-unseen performance is unstable. Report more seeds and provide confidence intervals. Discuss average performance rather than peak.

Response: We trained 3 additional seeds (789, 1011, 1213), yielding AUROCs of 0.657, 0.520, and 0.502 respectively. The 6-seed results are:

Estimate	AUROC	95% CI
3-seed mean	0.646 ± 0.124	[0.506, 0.745]
6-seed mean	0.603 ± 0.097	[0.532, 0.682]
Gradient Boosting	0.607	—

The 6-seed mean (0.603) is marginally *below* the gradient boosting baseline (0.607, bootstrap $p = 0.556$). We report this honestly and prominently:

“This result is important to report honestly: the original 3-seed mean (0.646) was moderately favorable; the 6-seed analysis demonstrates that DegradoMap’s advantage over hand-crafted feature baselines is not robust across all initializations.” (lines 621–622)

We explicitly recommend ensembling across ≥ 6 seeds for deployment (line 624) and note that individual seeds span a 0.25-AUROC range.

[Lines 619–625 (“Six-seed extended validation” paragraph); Table 1 lines 444–456; Figure 4 at line 632.]

M3: The lysine-related biological claim is not supported by ablation—removing the lysine indicator *improves* performance. Explain or retract.

Response: We now discuss this “lysine indicator paradox” directly in the Discussion with concrete evidence:

- With lysine indicator: val 0.584, test 0.477 (val-test gap: $-0.106 \rightarrow$ **overfitting**).
- Without lysine indicator: val 0.541, test 0.578 (val-test gap: $+0.038 \rightarrow$ generalizes).

This is a classic overfitting pattern: the lysine indicator improves validation but hurts test performance. We propose that the lysine-weighted *pooling mechanism* (Eq. 3) already encodes lysine information architecturally, making the binary node feature redundant and causing overfitting to training-set lysine distributions (lines 753–757).

We conclude: “architectural inductive biases (lysine pooling) are more effective than explicit feature annotation for this task,” and note that current feature design is suboptimal (Limitation 7, line 772).

[Lines 751–758 (“Lysine indicator paradox” paragraph); Figure 7 at line 762; Limitation 7 at line 772.]

M4: The E3 generalization claim should be narrower. Evidence supports mainly CRBN $\leftrightarrow$ VHL transfer.

Response: Agreed and revised throughout:

- Abstract: “CRBN $\rightarrow$ VHL E3-unseen transfer” (line 65).
- Section heading: “E3 generalization is limited to major ligases” (line 635).
- New per-E3 Table 5 (line 654) and Figure 5 (line 668) show: CRBN 0.758, VHL 0.396, cIAP1 0.417.
- VHL root cause analysis (lines 640–648): ruled out class imbalance (36.9% vs 38.4%) and dataset size (91 vs 125 proteins); identified structural heterogeneity as the likely cause.
- Conclusion: “CRBN $\rightarrow$ VHL transfer (0.811 AUROC)” (line 783).

[Lines 65, 635–648, 654–664 (Table 5), 668–672 (Figure 5), 783.]

M5: Validation setup should better align with the target-unseen setting, or explain why the current setup is sufficient.

Response: We now explicitly acknowledge this as a methodological limitation:

“Validation-split AUROC does not reliably predict target-unseen test AUROC. . . Ideally, we would use a target-unseen validation split for hyperparameter selection; however, this

would further reduce the already-limited training data (155 proteins $\rightarrow$ $\sim$ 120 training, $\sim$ 15 validation, $\sim$ 20 test).” (lines 327–328)

We state that the chosen hyperparameters may not be optimal for target-unseen generalization, and that this likely contributes to variance (line 330). This limitation is also listed as item 4 in the Limitations paragraph (line 769).

[Lines 326–330 (“Validation-test mismatch” paragraph in §3.5); Limitation 4 at line 769.]

Minor 1: Dataset limitations should be stated more directly in main results.

Response: A “Dataset characteristics and limitations” paragraph now appears at the start of Results, *before* any performance numbers (see response to Reviewer 9v2E, C1).

[Lines 431–436.]

Minor 2: The lack of external validation is an important limitation.

Response: Now stated as Limitation 6:

“No external validation: PROTAC-8K is the only publicly available benchmark with sufficient scale; generalization to data from other assay platforms or research groups is unvalidated.” (line 771)

[Line 771.]

Response to Reviewer PtZf (Rating: 3 — Borderline Reject)

W1: Performance is inconsistent across the paper: abstract reports 0.78 AUROC (95% CI 0.74–0.82), main results section reports 0.646 ± 0.124 , appendix reports 5-fold CV mean 0.565.

Response: We believe the “0.78” figure may refer to a prior draft; the current abstract reports 0.646 ± 0.124 (multi-seed average, line 65). To prevent any confusion, we have added a “Reconciling performance estimates” paragraph that explicitly maps each reported number to its evaluation protocol:

Number	Protocol	Role
0.603 ± 0.097	6-seed, single split	Most conservative
0.646 ± 0.124	3-seed, single split	Primary result
0.7449	Best seed	Peak (secondary)
0.657	Baseline (seed 42)	Single-seed reference
0.565 ± 0.052	5-fold CV	Conservative lower bound

The 6-seed mean is recommended for citation as the most conservative estimate.

[Lines 680–684 (“Reconciling performance estimates” paragraph in §4.3).]

W2: Robustness is limited. Target-unseen has high seed sensitivity, and 5-fold CV mean is substantially lower than the headline single-split number.

Response: We agree and have:

1. Extended to 6 seeds, yielding 0.603 ± 0.097 — marginally below gradient boosting (0.607); lines 619–625.
2. Stated explicitly that the improvement is **not statistically significant**: “bootstrap $p = 0.259$ ” (line 529) and “ $p = 0.556$ ” (line 621).
3. Recommended ensembling across ≥ 6 seeds for reliable deployment (line 624).
4. Reported both 3-seed and 6-seed results in Table 1 (lines 444–456) to show how the estimate changes with more seeds.

[Lines 444–456 (Table 1), 529–530, 619–625, 744–748 (“Honest performance assessment”).]

W3: Validation and model-selection protocol appears weak for the main use case. Validation AUROC does not reliably predict target-unseen test AUROC.

Response: Acknowledged as a methodological limitation (see response to Reviewer Y2rf, M5). We now discuss this explicitly in the Training Procedure section and Limitation 4.

[Lines 326–330 (“Validation-test mismatch”); Limitation 4 at line 769.]

W4: Generalization across E3 ligases remains limited. VHL performance is poor; rare ligases have too few samples.

Response: Fully addressed (see response to Reviewer Y2rf, M4). E3 claims are narrowed to CRBN $\leftrightarrow$ VHL transfer. A new per-E3 table (Table 5, line 654) and VHL root cause analysis (lines 640–648) are included. We now explicitly state:

“practitioners should expect substantially lower performance than the overall 0.646 mean when deploying on VHL-targeted proteins, and near-random performance for rare E3s.”
(lines 647–648)

[Lines 635–648, 654–664 (Table 5), 668–672 (Figure 5).]

W5: No external validation reduces confidence in real-world generalization.

Response: Stated as Limitation 6 (see response to Reviewer Y2rf, Minor 2). We caution against deployment without prospective validation on independently collected data.

[Line 771.]

Summary of All Changes

The table below maps each reviewer concern to the specific lines changed in the revised manuscript.

Concern	Reviewer(s)	Revision (lines in paper .tex)
Best seed over-emphasized	Y2rf M1, PtZf W1	Abstract L65; Table 1 L444–456; §4.2 L458–461; Conclusion L783–785
Target-unseen unstable	Y2rf M2, PtZf W2	§4.3 L619–625; Table 1 L450; Figure 4 L632; <i>p</i> -values L461, L529, L621
Lysine claim unsupported	Y2rf M3	§5 L751–758; Figure 7 L762; Limitation 7 L772
E3 claims too broad	Y2rf M4, PtZf W4	Abstract L65; §4.3 L635–648; Table 5 L654; Figure 5 L668; Conclusion L783
Validation misaligned	Y2rf M5, PtZf W3	§3.5 L326–330; Limitation 4 L769
Dataset limits buried	9v2E C1, Y2rf M6	§4.1 L431–436
No structural validation	9v2E C2	Limitation 5 L770
No case study	9v2E C3	§4.5 L686–703
Inconsistent reporting	PtZf W1	§4.3 L680–684 (reconciliation paragraph)
No external validation	Y2rf M7, PtZf W5	Limitation 6 L771
<i>New analyses added in revision:</i>		
6-seed validation	—	Table 1 L450; §4.3 L619–625; Figure 4 L632
Calibration analysis	—	§4.6 L705–732; Table 6 L714
Per-E3 breakdown	—	Table 5 L654; Figure 5 L668
VHL root cause	—	§4.3 L640–648
BRD4 case study	—	§4.5 L686–703
Lysine paradox figure	—	Figure 7 L762
Statistical significance	—	L461, L529, L621
Honest assessment	—	§5 L744–748